STUDENT PERSPECTIVES IN MILITARY GEOGRAPHY AND GEOSCIENCES

Navigating Environmental, Urban, and Security Challenges

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PREFACE

This edited volume, *Student Perspectives in Military Geography and Geosciences: Navigating Environmental, Urban, and Security Challenges*, emerged from an innovative student-centred initiative that reflects a strong culture of academic mentorship and collaborative research within Stellenbosch University's Faculty of Military Science. Most of the chapters are based on postgraduate research conducted in the Department of Military Geography at the Saldanha Campus. Additional contributions originate from the Departments of Physics (Mil) and Nautical Science. One chapter (Chapter 3) reflects collaborative undergraduate research undertaken by students from the Department of Geography and Environmental Studies at the main campus of Stellenbosch University, while Chapter 8 represents an undergraduate research project developed within the Faculty of Military Science. Many of the student contributions were developed through mentorship and co-authorship with senior academics from these departments, as well as from the Department of Geography at the University of South Africa (UNISA), highlighting the cross-institutional collaboration that underpins this volume.

Each chapter originated from undergraduate coursework or honours-level research conducted under structured academic supervision. Through this process of academic guidance and refinement, these student-led projects evolved into publishable scholarly contributions. Many chapters were further developed following their presentation at the 15th International Conference on Military Geosciences (ICMG 2024), held in Somerset West, South Africa, from 9 to 14 June 2024. Participation in this international forum provided student researchers with an opportunity to engage with global scholars, present their findings, and further strengthen their academic work. Collectively, the chapters presented in this volume demonstrate rigorous research, critical analysis, and the practical application of geospatial methods within contemporary military and environmental contexts. At a time when geospatial technologies are increasingly central to understanding environmental change, urban dynamics, and security challenges, this volume highlights the important role that emerging scholars can play in advancing applied research within military geography and geosciences.

All chapters in this volume underwent a formal academic review process in accordance with the policies of African Sun Media. Academic reviewers were independently selected, including two national reviewers from outside the authors' institutions and one international reviewer. The peer review followed a double-blind process: manuscripts were sent anonymously to reviewers, and their feedback was returned anonymously to the authors. Based on the reviewers' recommendations, the authors revised their work to ensure academic rigour and overall quality.

The book is thematically organised into four key sections, reflecting the diverse areas of focus within this interdisciplinary field.

The first section, *Environmental Management and Monitoring (Natural Environment)*, explores advanced geospatial methods for analysing environmental processes and risks. Chapters in this section include the assessment of radionuclide distribution in sediments (Chapters 1 and 4), the evaluation of land degradation in military training areas (Chapter 2), and the analysis of vegetation changes in conflict-affected regions (Chapter 3).

The second section, *Urban Dynamics, Crime, and Security (Human–Environmental Interactions)*, examines the spatial dimensions of urban development and security challenges. The chapters in this section explore residential expansion in informal settlements (Chapter 5), spatially targeted crime analysis in military and civilian contexts (Chapters 6 and 7), and the application of geospatial technologies to enhance human security strategies in the face of climate change, migration, and conflict (Chapter 8).

The third section, *Military, Maritime, and Security Infrastructure (Planning and Decision-Making)*, focuses on the strategic application of geospatial intelligence in infrastructure management and surveillance. Chapters 9, 10, and 11 provide insights into nuclear facility risk assessment (Chapter 9), utility infrastructure management in military contexts (Chapter 10), and maritime surveillance to detect illegal activities (Chapter 11).

The final section, *Sustainable and Renewable Energy Solutions*, addresses the growing importance of energy security and operational sustainability in defence contexts. Chapters 12, 13, and 14 examine photovoltaic solar farm site selection (Chapter 12), wave energy potential in coastal environments (Chapter 13), and the use of geospatial tools to support efficient military exercise planning (Chapter 14).

This volume showcases the academic capabilities of student researchers while highlighting the value of structured academic mentorship. By transforming coursework and honours research into peer-reviewed scholarly outputs, the contributors demonstrate how student-led research can meaningfully contribute to advancing knowledge in military geography and geosciences.

We commend the student authors and their academic mentors for their dedication, innovation, and scholarly commitment. We hope that this volume will encourage continued research, collaboration, and intellectual curiosity within the evolving fields of military geography and geosciences.

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Ivan Henrico, Susan Henrico, and Rikus le Roux

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